D3-Text Mining Methods

CIS 428

4/19/2026

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Introduction

This report applies multiple text mining techniques to analyze airline customer reviews from different analytical perspectives. Both supervised and unsupervised learning methods are used to extract meaningful insights from textual data. In the first section, supervised machine learning approaches are applied, while the later section focuses on unsupervised topic modeling.

Before applying these methods, the airline review dataset was preprocessed to ensure consistency and usability. The preprocessing steps included cleaning the text, removing noise, and transforming the data into a structured format suitable for analysis. The cleaned text was then converted into a Document-Term Matrix (DTM), which represents the frequency of words across all reviews and serves as the foundation for the applied models.

The classification methods used in this study include Naïve Bayes, K-Nearest Neighbors (KNN), and lexicon-based sentiment analysis, while Latent Dirichlet Allocation (LDA) was applied for topic modeling using clustering.

One of the primary research questions of this project is:
"What topics and themes appear most frequently in airline customer reviews?"

While topic modeling directly addresses this question, sentiment classification methods such as Naïve Bayes and KNN help provide additional insight by identifying how customer opinions (positive, neutral, or negative) are distributed across these themes. Together, these approaches provide a comprehensive understanding of both the structure and sentiment of customer feedback.

Classification Methods

In this section, supervised machine learning approaches, Naïve Bayes and K-Nearest Neighbors (KNN) are applied to classify customer reviews into sentiment categories: Positive, Neutral, and Negative. The objective is to evaluate how effectively different classification techniques can predict customer sentiment based on textual data.

Both models rely on a structured representation of text obtained through preprocessing and transformation into a Document-Term Matrix (DTM). A sample of DTM is shown in fig 1a. This representation allows textual data to be converted into numerical features, enabling machine learning algorithms to process and analyze the data.

Data Preparation

The airline review dataset was preprocessed using several standard text mining techniques to improve data quality and model performance. The text was first converted into a corpus and transformed to lowercase, followed by the removal of punctuation, numerical values, and common stop words such as "the," "and," and "is." A Document-Term Matrix (DTM) was then constructed, and sparse terms were removed to reduce noise and dimensionality. The resulting DTM was converted into a structured data frame suitable for modeling. The dataset was then split into 80% training data and 20% testing data, with Sentiment, categorized as Positive, Neutral, or Negative, used as the target variable for classification.

Naïve Bayes Classification

Method Overview

Naïve Bayes is a probabilistic classification algorithm based on Bayes' Theorem. It assumes that the features (words) are conditionally independent given the class label. Despite this simplifying assumption, it is highly effective for text classification tasks due to its ability to handle high-dimensional data efficiently.

Model Implementation

The Naïve Bayes model was trained using the Document-Term Matrix derived from the preprocessed dataset. A binary representation of word presence was used to improve model stability and reduce the impact of highly frequent terms. Laplace smoothing was applied to handle zero-frequency issues and ensure that unseen words in the test data do not result in zero probabilities.

Evaluation and Results

The performance of the Naïve Bayes model was evaluated using a confusion matrix, as shown in Fig. 1b. The confusion matrix compares predicted sentiment labels with actual labels from the test dataset, providing insight into classification accuracy and error distribution. The inclusion of Laplace smoothing improved the robustness of the model by addressing sparse data issues.

Discussion

The Naïve Bayes classifier performed effectively in predicting customer sentiment. It demonstrated strong performance in identifying dominant sentiment classes, particularly positive and negative reviews. However, due to its independence assumption, it may not fully capture contextual relationships between words, which can affect performance in more nuanced cases.

K-Nearest Neighbors (KNN) Classification

Method Overview

K-Nearest Neighbors (KNN) is a non-parametric, instance-based learning algorithm that classifies data points based on the majority class of their nearest neighbors in the feature space. It uses distance metrics to determine similarity between observations.

Model Implementation

The KNN model was applied to the same Document-Term Matrix used for Naïve Bayes. Since KNN is sensitive to feature scaling, normalization was performed before model training. The model was trained using the training dataset, and classification was performed using k = 5, where each review is assigned a sentiment based on the majority class among its five nearest neighbors.

Evaluation and Results

The confusion matrix for KNN with k = 5 is presented in Fig. 2a, showing the model's classification performance across sentiment categories. Additionally, model performance was evaluated for multiple values of k, and the resulting accuracy comparison is shown in Fig. 2b. This analysis helps determine the optimal value of k and assess model stability.

Discussion

KNN demonstrated the ability to classify sentiment based on similarity in textual patterns. The results indicate that model performance varies depending on the choice of k, as shown in Fig. 2b. While KNN provides an intuitive approach to classification, it is computationally more expensive than Naïve Bayes and can be affected by the high dimensionality of text data.

Lexicon-Based Sentiment Analysis

Method Overview

In addition to supervised machine learning approaches, a lexicon-based sentiment analysis was conducted to evaluate customer sentiment. This method does not require labeled training data and instead relies on predefined sentiment dictionaries to assign sentiment scores to words.

Implementation

The dataset was transformed into a tidy format, where each review was tokenized into individual words and stop words were removed to focus on more meaningful terms. The remaining words were then matched with established sentiment lexicons, including Bing, AFINN, and NRC. The Bing lexicon classifies words as either positive or negative, the AFINN lexicon assigns numerical sentiment scores ranging from -5 to +5, and the NRC lexicon categorizes words into different emotional and sentiment-based groups.

Results

A word cloud was generated to visualize the most frequently occurring terms in the dataset, as shown in Fig. 3a. This visualization highlights dominant words such as “flight”, “service”, and “seat”, indicating that these aspects are central to customer feedback.

Additionally, a sentiment comparison plot across the three lexicons is presented in Fig. 3b, illustrating differences in how each lexicon evaluates sentiment distribution.

Discussion

The lexicon-based approach provided a quick and interpretable overview of sentiment within the dataset. The word cloud and sentiment comparison reveal that operational aspects of the airline experience are frequently discussed. However, this method does not account for context, sarcasm, or domain-specific language, which may limit its accuracy compared to machine learning models.

Topic Modeling using LDA

Method Overview

Latent Dirichlet Allocation (LDA) is an unsupervised learning technique used to identify hidden topics within a collection of documents. It groups words into topics based on their co-occurrence patterns across the dataset.

Implementation

The preprocessed dataset was converted into a Document-Term Matrix, and sparse terms were removed to improve model performance. The LDA model was trained with k = 3 topics, and the most significant terms for each topic were extracted.

Results

The top contributing terms for each topic are shown in Figure 4, and these terms were used to interpret the main themes in the dataset. Topic 1 represents overall flight experience and service quality, Topic 2 focuses on airline operations and business class experience, and Topic 3 reflects airport experience, seating, and travel time. These groupings show that the topics are clearly separated based on different parts of the travel experience.

Overall, the results suggest that customer reviews mainly focus on service quality, airline operations, and the overall travel experience. This indicates that passengers evaluate multiple aspects of their journey, including both in-flight service and factors related to the airport and travel process, rather than focusing on just one area.

Discussion

The LDA model successfully identified meaningful themes within the dataset, directly addressing the research question. Some common words, such as “flight”, appeared across multiple topics, reflecting their general importance. Overall, topic modeling provided valuable insights into the structure of customer feedback without requiring labeled data.

Conclusion:

This study applied multiple text mining techniques, including Naive Bayes, K-Nearest Neighbors (KNN), lexicon-based sentiment analysis, and Latent Dirichlet Allocation (LDA), to analyze airline customer reviews. The classification models, Naive Bayes and KNN, were used as predictive approaches to determine customer sentiment based on textual features. Among these, Naive Bayes demonstrated strong performance. The lexicon-based method offered a simple and interpretable way to assess sentiment without requiring labeled data. LDA topic modeling was applied as a clustering technique. The model successfully identified key themes within the dataset, including overall service quality, business class experience, and airport-related factors. These findings demonstrate that customer reviews primarily focus on operational service aspects and travel experience. Overall, combining classification and clustering methods provided a comprehensive understanding of both customer sentiment and underlying themes. While supervised models offered predictive capabilities, unsupervised topic modeling revealed deeper insights into the structure of customer feedback.

Results:

```
> inspect(dtm)
<<DocumentTermMatrix (documents: 3701, terms: 14218)>>
Non-/sparse entries: 250767/52370051
Sparsity           : 100%
Maximal term length: 33
Weighting          : term frequency (tf)
Sample             :
      Terms
Docs   airways british crew flight food good london seat seats service
  1115       0       0    6      3    5    2      2    1     0      11
  1195       0       0    1      8    0    0      1    7     4       0
  1380       0       1    3      2    0    4      1    4     1       6
  1384       2       2    1      2    2    6      1    0     2       2
  1451       0       0    5      5    2    1      2    0     1      11
  1759       0       0    3      3    3    4      2    0     0       4
```

Fig 1a. Sample of a Document-Term Matrix showing the word-frequency representation of reviews.

```
   Cell Contents
|-------------------------|
|                       N |
|           N / Col Total |
|-------------------------|


Total Observations in Table:  740


             | Actual
   Predicted |  Negative |   Neutral |  Positive | Row Total |
-------------|-----------|-----------|-----------|-----------|
    Negative |       135 |        31 |        14 |       180 |
             |     0.547 |     0.279 |     0.037 |           |
-------------|-----------|-----------|-----------|-----------|
     Neutral |        48 |        30 |        41 |       119 |
             |     0.194 |     0.270 |     0.107 |           |
-------------|-----------|-----------|-----------|-----------|
    Positive |        64 |        50 |       327 |       441 |
             |     0.259 |     0.450 |     0.856 |           |
-------------|-----------|-----------|-----------|-----------|
Column Total |       247 |       111 |       382 |       740 |
             |     0.334 |     0.150 |     0.516 |           |
-------------|-----------|-----------|-----------|-----------|
```

```
   Cell Contents
|-------------------------|
|                       N |
|           N / Col Total |
|-------------------------|


Total Observations in Table:  740


             | Actual
   Predicted |  Negative |   Neutral |  Positive | Row Total |
-------------|-----------|-----------|-----------|-----------|
     Neutral |       247 |       111 |       381 |       739 |
             |     1.000 |     1.000 |     0.997 |           |
-------------|-----------|-----------|-----------|-----------|
    Positive |         0 |         0 |         1 |         1 |
             |     0.000 |     0.000 |     0.003 |           |
-------------|-----------|-----------|-----------|-----------|
Column Total |       247 |       111 |       382 |       740 |
             |     0.334 |     0.150 |     0.516 |           |
```

Figures 1b. Confusion matrix for Naive Bayes classification showing prediction performance on test data. The above figure shows the accuracy of 66.4%. The figure below is, with Laplace smoothing, but the prediction doesn't get better.

```
[1] "Confusion Matrix for k=5:"
          Actual
Predicted  Negative Neutral Positive
  Negative      205      25       29
  Neutral         0       2        2
  Positive      184      58      233
Model Accuracy (k=5): 59.62 %
```

Fig 2a. Confusion matrix for KNN classification using k = 5.

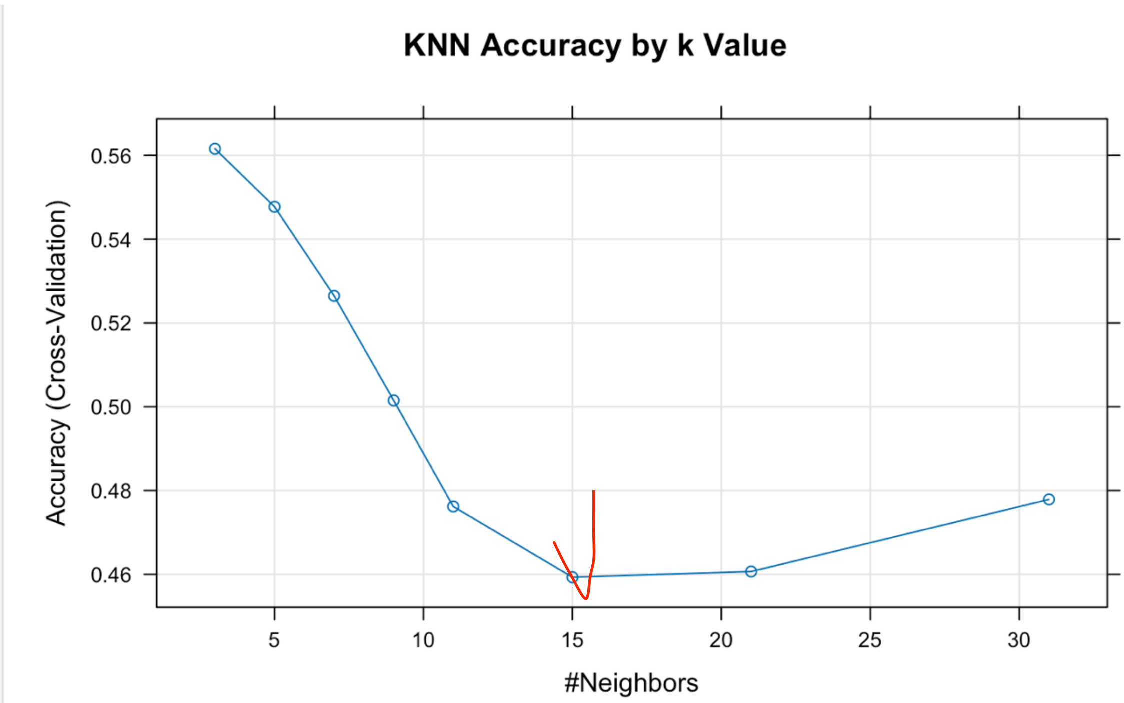

Fig 2b. Model performance comparison across different values of k in KNN

```
[1] "Confusion Matrix for Best Model:"
          Actual
Predicted  Negative Neutral Positive
  Negative      234      35       48
  Neutral         8       2        6
  Positive      147      48      210
```

Fig 2c. Confusion Matrix for best model. (delete)

Fig 3a. Word cloud showing the most frequently occurring terms in airline reviews.

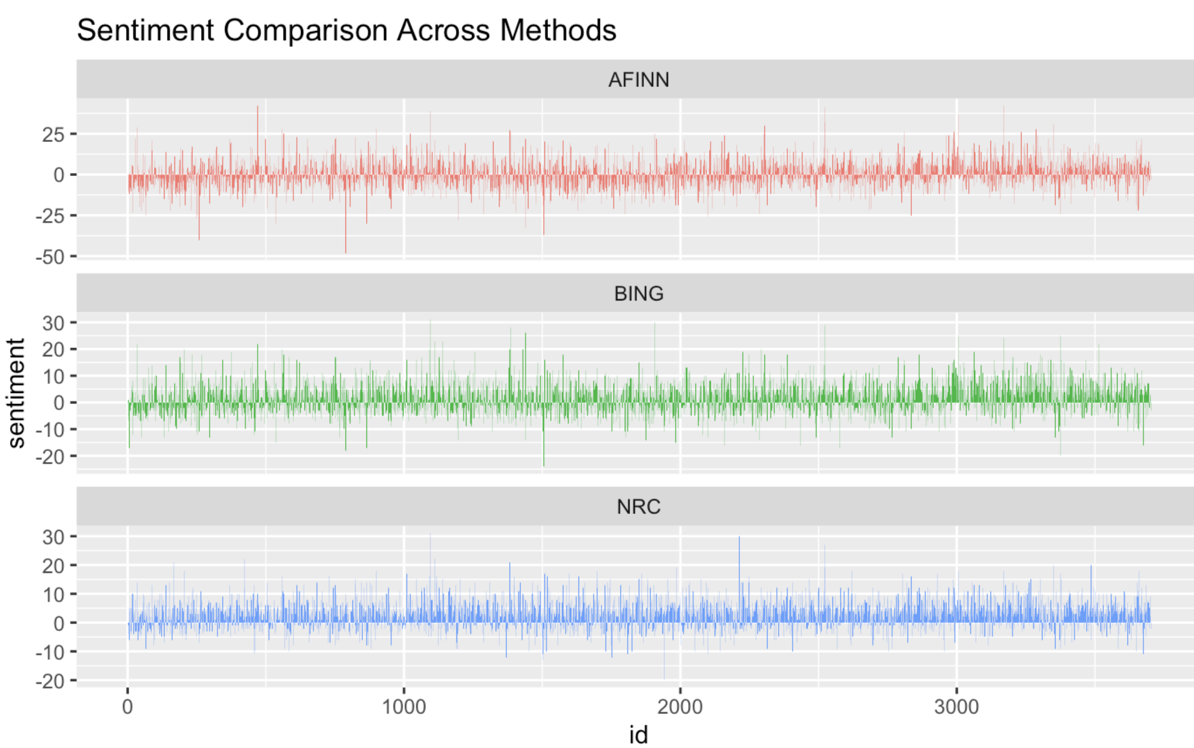

Fig 3b. Distribution of sentiment words across 3 methods.

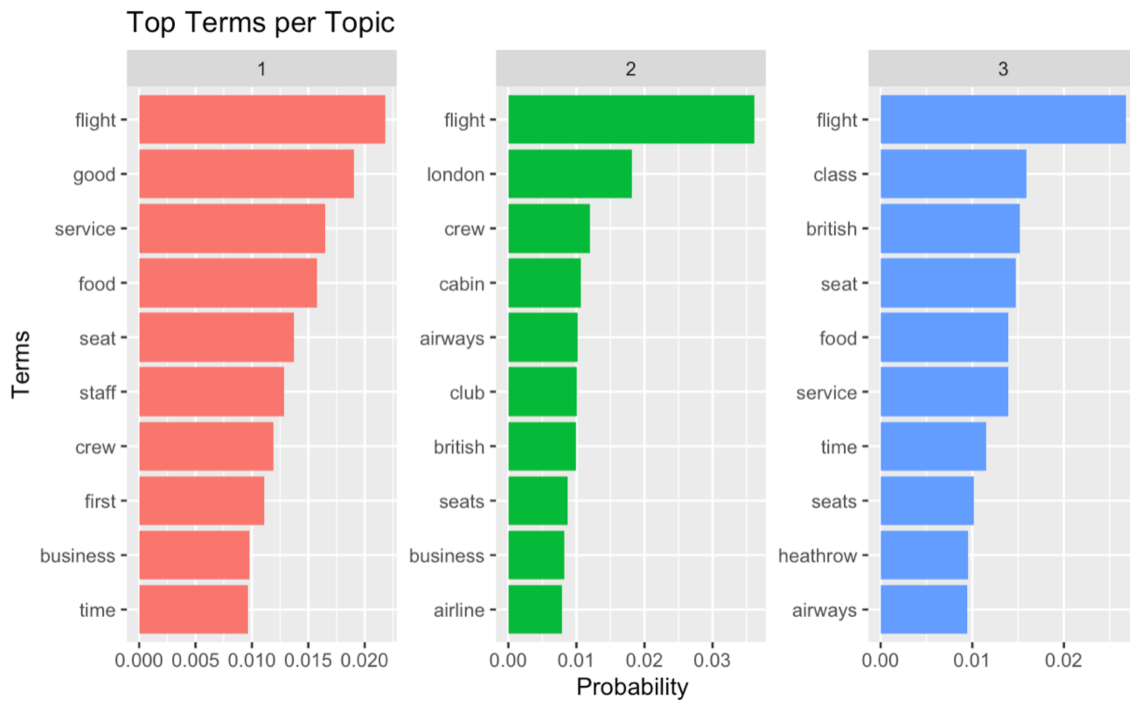

Fig 4. Top contributing terms for each topic identified using LDA.